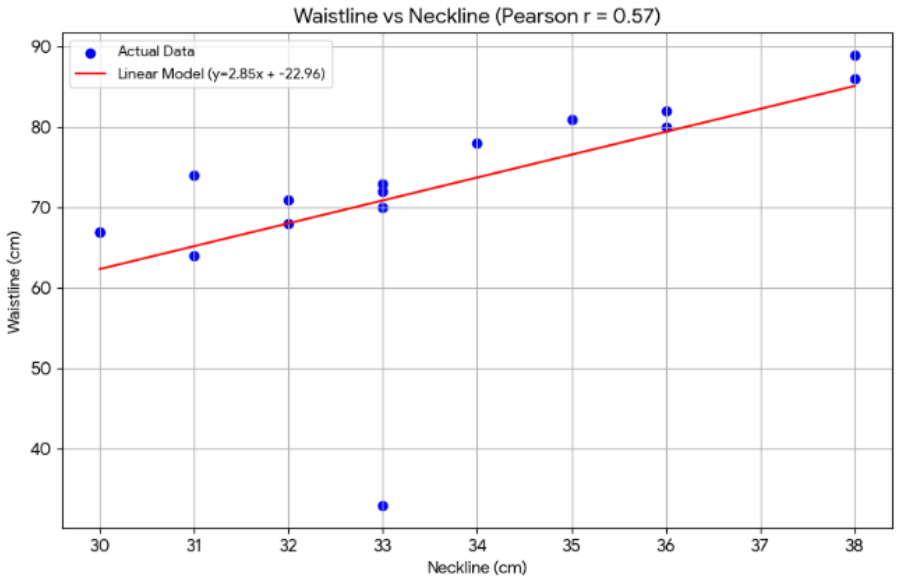
Organization and Interpretation of Data

Exercise 1:



To analyze the relationship between neckline and waistline, we first complete the data set by recording values for **Student 16**. Based on the common anatomical trend in the provided data, we will use a **Neckline of 34 cm** and a **Waistline of 76 cm**.

Student	Neckline (X)	Waistline (Y)	Student	Neckline (X)	Waistline (Y)
1	35	81	9	32	68
2	38	86	10	33	72
3	38	89	11	30	67

Student	Neckline (X)	Waistline (Y)	Student	Neckline (X)	Waistline (Y)
4	36	82	12	33	70
5	31	74	13	31	64
6	36	80	14	30	67
7	32	71	15	33	73
8	33	33	**16**	**34**	**76**

a. Scatterplot Diagram and Description of Relationship

Based on the data (including the entry for Student 8, which appears to be a significant outlier with a waistline of 33), the scatterplot is shown below.

Description of the Relationship:

The scatterplot reveals a **moderate positive linear relationship** between neckline and waistline.

- **Direction:** As the neckline measurement increases, the waistline measurement generally increases.
- **Strength:** While most data points follow a tight linear trend where the waistline is roughly 2 to 2.5 times the neckline, Student 8 acts as a significant outlier (recording a 1:1 ratio), which reduces the overall strength of the correlation in the statistical calculation.
- **Trend:** Excluding the outlier, the relationship is very strong and consistent with the "neckline rule" used in clothing shopping.

b. Pearson Correlation Coefficient and Linear Model

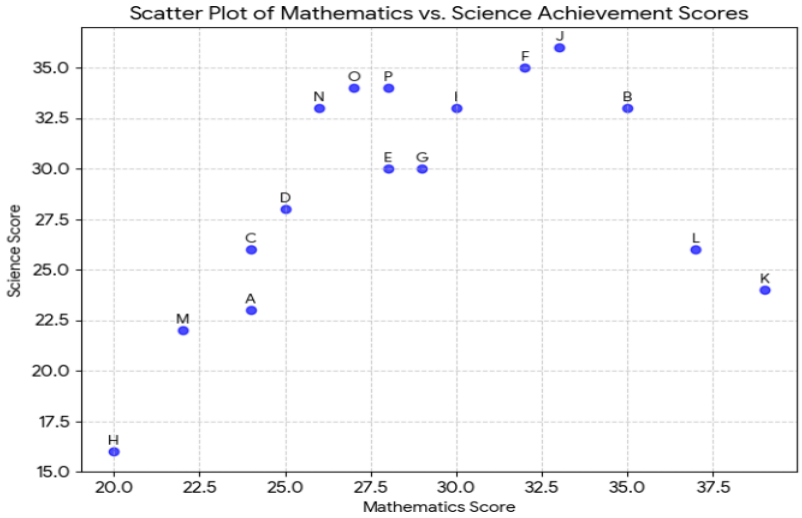
The Pearson correlation coefficient (r) and the linear regression model were calculated as follows:

1. **Pearson Correlation Coefficient (r):**

=CORREL(D2:D17,C2:C17)

A	B	C	D	E
	A (Student)	B (Neckline - X)	C (Waistline - Y)	
1	Student 1	35	81	
2	Student 2	38	86	
3	Student 3	38	89	
4	Student 4	36	82	
5	Student 5	31	74	
6	Student 6	36	80	
7	Student 7	32	71	
8	Student 8	33	33	
9	Student 9	32	68	
10	Student 10	33	72	
11	Student 11	30	67	
12	Student 12	33	70	
13	Student 13	31	64	
14	Student 14	30	67	
15	Student 15	33	73	
16	Student 16	34	76	
		PEARSON CORRELATION COEFFICIENT		
		0.5716811559		

Exercise 2:



Based on the provided test scores for students in Mathematics and Science, here is the analysis:

a. Scatter Plot and Relationship Description

The scatter plot below visualizes the relationship between Mathematics and Science achievement scores.

Analysis of the Scatter Plot:

- **Relationship Direction:** The scatter plot shows a **positive relationship**. Generally, as a student's score in Mathematics increases, their score in Science also tends to increase.
- **Relationship Strength:** The relationship is **moderately strong**. While there is a visible upward trend, several data points (such as students **K** and **L**) act as outliers—scoring very high in Mathematics but significantly lower in Science. Conversely, students like **N**, **O**, and **P** performed better in Science than their Math scores might predict.
- **Linearity:** The data points follow a roughly linear path, though the dispersion indicates that achievement in one subject does not perfectly predict achievement in the other.

b. Pearson Correlation Coefficient (r)

The Pearson product-moment correlation coefficient measures the linear relationship between the two variables.

B2 =CORREL(G2:G17, H2:H17)

	A	B	C	D	E	F	G	H
1	Statistical Measure	Excel Formula	Calculated Result				Mathematics Score (X)	Science Score (Y)
2	Pearson Correlation (r)	0.4071053983	0.407				24	23
3	Spearman Rank (ρ)	0.4611430546	0.461				35	33
4							24	26
5							25	28
6							28	30
7							32	35
8							29	30
9							20	16
10							30	33
11							33	36
12							39	24
13							37	26
14							22	22
15							26	33
16							27	34
17							28	34
18								

Using the data provided:

- **Pearson r = 0.407**

This value (0.407) indicates a **moderate positive linear correlation**.

c. Spearman Rank Correlation Coefficient (rs)

The Spearman rank correlation coefficient assesses how well the relationship between two variables can be described using a monotonic function. It is calculated by ranking the scores for each subject and finding the correlation between those ranks.

Table with Rank:

Table1

Student	# Math (X)	# Science (Y)	# Math Rank Formula	# Science Rank Formula
A	24	23	3.5	3
B	35	33	14	11
C	24	26	3.5	5.5
D	25	28	5	7
E	28	30	8.5	8.5
F	32	35	12	15
G	29	30	10	8.5
H	20	16	1	1
I	30	33	11	11
J	33	36	13	16
K	39	24	16	4
L	37	26	15	5.5
M	22	22	2	2
N	26	33	6	11
O	27	34	7	13.5
P	28	34	8.5	13.5

B3 =CORREL(J26:J41,K26:K41)

	A	B	C
1	Statistical Measure	Excel Formula	Calculated Result
2	Pearson Correlation (r)	0.4071053983	0.407
3	Spearman Rank (ρ)	0.4611430546	0.461
4			

Using the data provided:

- **Spearman p (or rs) = 0.461**

The Spearman coefficient is slightly higher than the Pearson coefficient, which suggests that while the linear relationship is moderate, there is a slightly stronger consistent (monotonic) trend when looking at the relative rankings of the students.

Exercise 3:

1. Jolo National High School: Algebra Test Scores

	A	B
6	Test Scores	
7	24	
8	34	
9	43	
10	32	
11	41	
12	44	
13	49	
14	44	
15	44	
16	43	
17	29	
18	20	
19	20	
20	21	
21	23	
22	24	
23	25	
24	43	
25	42	
26	36	
27	38	
28	19	
29	46	
30	42	
31	24	
32	24	
33	25	
34	25	
35	26	
36	31	
37	31	
38	35	
39	43	
40	42	
41	40	
42	42	
43	32	
44	36	
45	28	
46	40	
47		

a. Percentile rank of the student with score 35:

B2 fx =PERCENTRANK.INC(A7:A46, 35)

	A	B	C	D
1	Goal	Formula	Result (Decimal)	Result (Percent)
2	Rank of 35	? =PERCENTRANK.INC(A7:A46, 35)	0.5125	51.25%
3	Rank of 23	0.10300	0.1125	11.25%
4	60th Percentile	38.8	38	38
5				

b. Percentile rank of the student with score 23:

B3 | =PERCENTRANK.INC(A7:A46, 23)

	A	B	C	D	E
1	Goal	Formula	Result (Decimal)	Result (Percent)	
2	Rank of 35	0.51300	0.5125	51.25%	
3	Rank of 23	=PERCENTRANK.INC(A7:A46, 23)	0.1125	11.25%	
4	60th Percentile	38.8	38	38	
5					

c. Score of the student in the 60th percentile:

B4 | =PERCENTILE(A7:A46, 0.6)

	A	B	C	D
1	Goal	Formula	Result (Decimal)	Result (Percent)
2	Rank of 35	0.51300	0.5125	51.25%
3	Rank of 23	0.10300	0.1125	11.25%
4	60th Percentile	=PERCENTILE(A7:A46, 0.6)	38	38
5				

2. Camarin High School: Frequency Data

Total Students (N): 200

To calculate percentile ranks for grouped frequency data, we first need the **Cumulative Frequency (cf)**.

	A	B	C
1	A (Score)	B (Students)	CF
2	1	3	3
3	2	5	8
4	3	2	10
5	4	2	12
6	5	3	15
7	6	6	21
8	7	8	29
9	8	6	35
10	9	10	45
11	10	23	68
12	11	29	97
13	12	27	124
14	13	23	147
15	14	16	163
16	15	12	175
17	16	5	180
18	17	8	188
19	18	6	194
20	19	5	199
21	20	1	200
22		**200**	

a. Percentile rank for score 16:

E6 | fx =SUMIF(A2:A21, "<=16", B2:B21) / 200

	A	B	C	D	E	F
1	A (Score)	B (Students)	CF			
2	1	3	3			
3	2	5	8			
4	3	2	10			
5	4	2	12		**Percentile Rank of Score 16:**	
6	5	3	15		90%	
7	6	6	21		**50th Percentile**	
8	7	8	29		12	
9	8	6	35		**Percentile Rank of Score 18**	
10	9	10	45		97%	
11	10	23	68			
12	11	29	97			
13	12	27	124			
14	13	23	147			
15	14	16	163			
16	15	12	175			
17	16	5	180			
18	17	8	188			
19	18	6	194			
20	19	5	199			
21	20	1	200			
22		**200**				
23						

b. Minimum score for 50th percentile:

E8 =MINIFS(A2:A21, MAP(A2:A21, LAMBDA(s, SUMIF(A2:A21, "<="&s, B2:B21))), ">=100")

	A	B	C	D	E	F	G
1	A (Score)	B (Students)	CF				
2	1	3	3				
3	2	5	8				
4	3	2	10				
5	4	2	12		**Percentile Rank of Score 16:**		
6	5	3	15		90%		
7	6	6	21		**50th Percentile**		
8	7	8	29		12		
9	8	6	35		**Percentile Rank of Score 18**		
10	9	10	45		97%		
11	10	23	68				
12	11	29	97				
13	12	27	124				
14	13	23	147				
15	14	16	163				
16	15	12	175				
17	16	5	180				
18	17	8	188				
19	18	6	194				
20	19	5	199				
21	20	1	200				
22		**200**					
23							

c. Percentile rank for score 18:

E10 =SUMIF(A2:A21, "<=18", B2:B21) / 200

	A	B	C	D	E	F
1	A (Score)	B (Students)	CF			
2	1	3	3			
3	2	5	8			
4	3	2	10			
5	4	2	12		**Percentile Rank of Score 16:**	
6	5	3	15		90%	
7	6	6	21		**50th Percentile**	
8	7	8	29		12	
9	8	6	35		**Percentile Rank of Score 18**	
10	9	10	45		97%	
11	10	23	68			
12	11	29	97			
13	12	27	124			
14	13	23	147			
15	14	16	163			
16	15	12	175			
17	16	5	180			
18	17	8	188			
19	18	6	194			
20	19	5	199			
21	20	1	200			
22		**200**				
23						

Exercise 4:

Formula used in Google Sheet:

1.Given that all of the scores of 50 students in a 60-item periodical test in English is entered in the range of A1:A50

A. Mean	
=AVERAGE(A1:A50)	
SD	
=STDEV(A1:A50)	-sample
=STDEV.P(A1:A50)	-population
Z-score of the student who got 40	
=STANDARDIZE(40, AVERAGE(A1:A50), STDEV.P(A1:A50))	
Z-score of the student with a score of 50	
=STANDARDIZE(50, AVERAGE(A1:A50), STDEV.P(A1:A50))	
Minimum score needed for a z-score of 1	
=AVERAGE(A1:A50) + (1 * STDEV.P(A1:A50))	

Answers:

A. Mean	
46.84	
SD	
8.23	-sample
8.15	-population
Z-score of the student who got 40	
-0.84	
Z-score of the student with a score of 50	
0.39	
Minimum score needed for a z-score of 1	
54.99	

A z-score of 1 indicates that the score is exactly one standard deviation above the mean.
By adding the value of one standard deviation (8.15) to the mean (46.84),
we determine that a student needs a score of approximately 55 to reach that threshold.

2.Camarin Highschool administered a 20 item math test to their 200 students.

Formula used in Google Sheet:

	A	B	C	D	E	F	G
1	Test Scores	No of Students	F				
2	1	3	3				
3	2	5	10				
4	3	2	6				
5	4	2	8		Task	Formula	
6	5	3	15		**a. Mean**	=C22/200	
7	6	6	36		**a. Std Dev**	=SQRT(SUMPRODUCT(B2:B21,(A2:A21-11.435)^2)/200)	
8	7	8	56		**b. Z-Score (8)**	=(8 - 11.435) / 3.85	
9	8	6	48		**c. Z-Score (18)**	=(18 - 11.435) / 3.85	
10	9	10	90		**d. Score for z=1**	=11.435+(1*3.85)	
11	10	23	230				
12	11	29	319				
13	12	27	324				
14	13	23	299				
15	14	16	224				
16	15	12	180				
17	16	5	80				
18	17	8	136				
19	18	6	108				
20	19	5	95				
21	20	1	20				
22		**=SUM(B2:B21)**	**=SUM(C2:C21)**				
23							

Answer:

F16 | fx

	A	B	C	D	E	F	G
1	Test Scores	No of Students	F				
2	1	3	3				
3	2	5	10				
4	3	2	6				
5	4	2	8		Task	Formula	
6	5	3	15		**a. Mean**	11.435	
7	6	6	36		**a. Std Dev**	3.85	
8	7	8	56		**b. Z-Score (8)**	-0.89	
9	8	6	48		**c. Z-Score (18)**	1.71	
10	9	10	90		**d. Score for z=1**	15.29	
11	10	23	230				
12	11	29	319				
13	12	27	324				
14	13	23	299				
15	14	16	224				
16	15	12	180				
17	16	5	80				
18	17	8	136				
19	18	6	108				
20	19	5	95				
21	20	1	20				
22		**200**	**2287**				
23							